

Alex Rivers

Systems & Infrastructure Engineer (ML focus)

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Computer Science student specializing in distributed systems, storage engines, and ML training infrastructure. Experienced in building transactional key-value databases, implementing consensus algorithms, and optimizing high-throughput data processing pipelines. Passionate about kernel development, low-latency network programming, and platform automation.

Professional Experience

Systems & Linux Kernel Intern

May 2025 -- Aug 2025

KernelCorp Technologies

San Jose, CA (Remote)

- Designed and implemented an eBPF/XDP-based load-balancer that reduced packet processing latency by 35% compared to IPVS-based routing.
- Patched network driver queues to optimize cache-line utilization, resulting in a 12% increase in packet throughput for 100GbE interfaces.
- Developed a Linux Kernel Module (LKM) for virtual device drivers to mock high-frequency sensors for CI/CD test automation.
- Collaborated with upstream developers to patch a race condition in kernel slab memory allocation under heavy network load.

Distributed Storage Intern

May 2024 -- Aug 2024

StoreScale Systems

Seattle, WA

- Integrated RocksDB as a pluggable storage engine for a custom Raft-based transactional database, improving write throughput by 40%.
- Engineered an asynchronous log-compaction pipeline in Go to reduce tail latency of client writes under disk I/O bottlenecks.
- Built a cluster simulation framework to perform partition testing (Jepsen-style) and validated correctness of linearizable reads.
- Optimized gRPC request-serialization paths using Protobuf custom-marshaling, achieving an 18% reduction in CPU utilization.

Projects

TensorFlow & PyTorch ML Training Orchestrator (Python, PyTorch, Kubernetes, CUDA, gRPC, Docker)

Mar 2025

- Developed a custom Kubernetes controller in Go that schedules distributed PyTorch training jobs across multiple GPU-enabled nodes.
- Configured torch.distributed tensor-parallel setups with custom communication primitives, optimizing NCCL ring-allreduce performance.
- Implemented a dynamic fault-recovery agent that restarts failed worker pods, restoring from latest weights checkpoints in under 30 seconds.
- Designed an active GPU allocation scheduler that schedules training jobs to maximize NVLink interconnect bandwidth between GPUs.

Technical Skills

Tools: Git & GitHub Actions, Prometheus & Grafana
ML & AI Infrastructure: PyTorch, CUDA & NCCL, Triton Inference Server
Languages: Rust, C++, Go, Python, C
Systems & Infrastructure: Kubernetes, Docker

Education

Bachelor of Science

Tech Institute of Technology

Sept 2023 -- May 2027

Boston, MA

GPA: 3.95/4.00

- Teaching Assistant: Distributed Systems (CS 452), Operating Systems (CS 321)
- Coursework: Distributed Systems, Database Internals, Compiler Construction, Computer Architecture, Advanced Networks